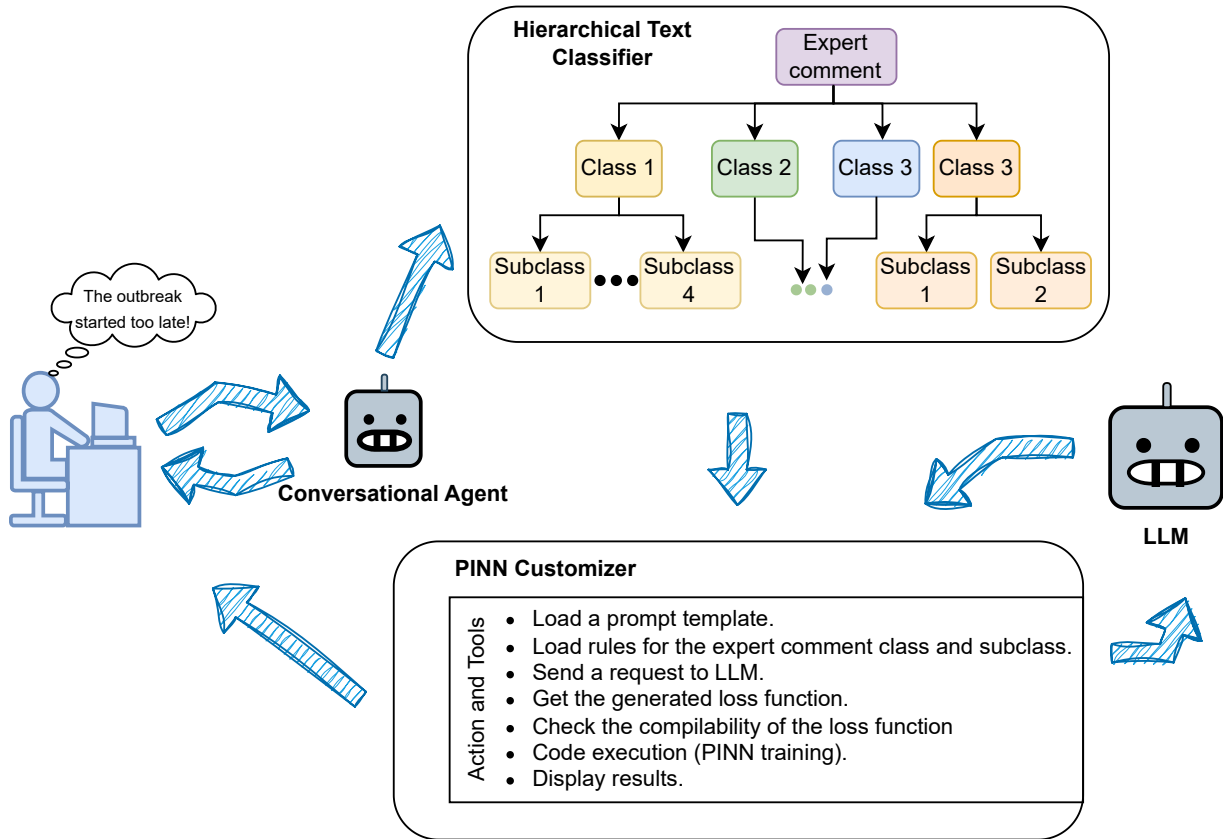
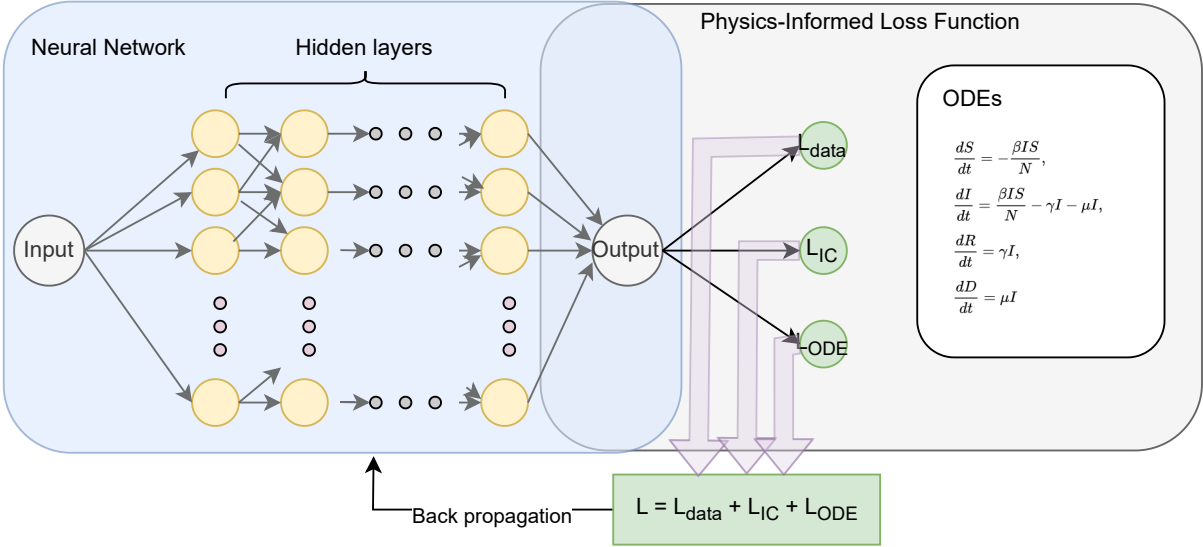




Physics-informed neural network



Prompt

Task Information

You are a machine learning expert specializing in Physics-Informed Neural Networks (PINNs) and epidemiological modeling. Your task is to modify a loss function for a SIRD model to better align with epidemiological dynamics ...

Primary loss function

```
def loss_function(S_hat, S_pred, I_hat, I_pred, D_hat, ...
```

Expert comment

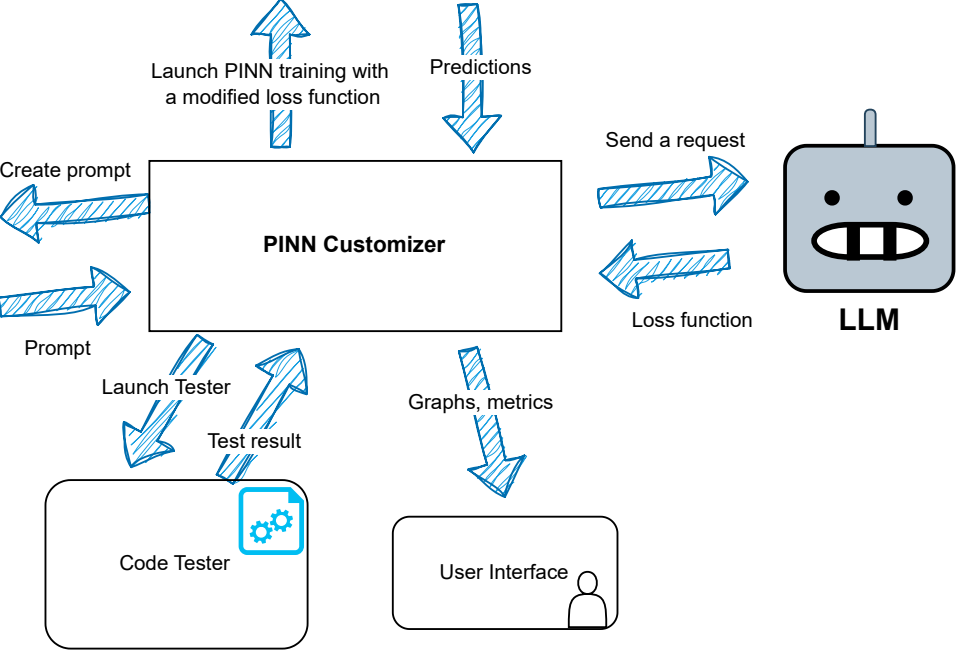
The graph of the number of infected people shows too sharp a decline after the peak – in reality, the 'tail' of the epidemic should be longer.

Expert comment class and subclass information

The error is due to the fact that the behavior of the epidemic curve...

Rules

Add penalty to term2, taking into account expert's comment. Changing the rest of the function is PROHIBITED...



Expert comment

1. Behavior of the
epidemiological curve
(visual characteristics)

1.2 Graph of the
number of people
susceptible to the
disease

1.1 Graph of the
number of infected

1.3 Graph of the
number of recovered
patients

1.4 Graph of the
number of deaths

2. Taking into account
measures to counter the
spread of the epidemic

2.1 Temporal
mismatch of the effect
of measures

2.2 Incorrect
reflection of the
effects of measures

2.3 Lack of reflection
of the effects of
measures

3. Parameters or temporal dynamics
of the epidemiological process
(numerical characteristics)

3.1 Time discrepancy

3.2 Discrepancy in
quantity

4. Position of the epidemic
peak

4.1 Early/Late Peak

4.2 Incorrect number
of cases during the
peak

